PDE-Agents: An LLM-Orchestrated Multi-Agent Framework for Automated Finite Element Simulations with Knowledge Graph-Augmented Reasoning

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Abstract

We present *PDE-Agents*, a multi-agent ecosystem that automates the full lifecycle of partial differential equation (PDE) / finite element method (FEM) simulations through natural-language interaction. The system combines three specialist large language model (LLM) agents—Simulation, Analytics, and Database—orchestrated via a LangGraph supervisor graph, with a locally-deployed open-source LLM stack (Qwen3-Coder-Next, Llama 4 Scout) running on dual NVIDIA RTX PRO 6000 Blackwell GPUs (≈196 GB combined VRAM). The architecture is model-agnostic; we report cross-model validation across two generations of open-source LLMs. A central novelty is the integration of a *GraphRAG* knowledge base (Neo4j + 768-dimensional vector embeddings) that enriches agents with curated material properties, known failure patterns, and prior run lineage. We report six empirical contributions: (i) a formal verification and validation (V&V) study confirming second-order spatial convergence ($\mathcal{O}(h^2)$) for all three benchmark cases on the heat equation solver; (ii) a three-way ablation study across 50 benchmark tasks with a frozen KG, comparing KG On, KG Off, and KG Smart, showing that while KG Off achieves 100% task completion, both KG-enabled modes produce *higher-quality* outputs (KG Smart: physics score 0.918 vs. 0.853 for KG Off, MPF 0.898 vs. 0.796), and that our *KG Smart* integration—combining warm-start injection with lazy conditional retrieval—achieves 94% success while maximising output fidelity; (iii) a novel-material experiment using three fictional materials whose properties exist only in the KG, where KG-enabled modes achieve perfect material property fidelity (MPF = 1.00) versus 0.34 for the KG-free baseline; (iv) an adaptive KG decision framework (Algorithm 1) that selects the optimal retrieval mode per task; (v) production-scale agent quality metrics from 1,369 real simulation runs showing 97.8% overall success and a 57.6% first-try rate; and (vi) a controlled 100-task KG growth experiment showing that accumulated run history yields a difficulty-dependent quality gain, with hard-task MPF improving by 8.8% between passes while easy and novel tasks remain at ceiling. All code, models, and evaluation artifacts are released openly. Our findings demonstrate that *integration pattern*, not knowledge content, determines whether GraphRAG augmentation helps or hinders LLM agents, and identify concrete design principles for autonomous FEM simulation assistants.

Keywords: Large language model agents; finite element method; knowledge graph; GraphRAG; multi-agent systems; scientific computing automation; verification and validation.

1 Introduction

Finite element method simulations are central to engineering analysis across structural mechanics, heat transfer, fluid dynamics, and electromagnetics. Yet the gap between a domain expert's intent—"what if I use aluminium instead of steel?"—and a running, verified simulation involves a sequence of error-prone manual steps: geometry creation, boundary condition specification, solver parameter selection, result interpretation, and iterative debugging.

The recent emergence of *reasoning-capable* LLMs [1–3] has opened the possibility of agents that can autonomously traverse this workflow. Prior work on "AI for scientific computing" has largely focused on surrogate modelling [4–6], operator learning [7], or dataset-specific fine-tuning [8]. Much less attention has been paid to *procedural automation*—using LLMs to *set up and manage* solvers rather than to replace them.

Meanwhile, the *retrieval-augmented generation* (RAG) paradigm [9] has demonstrated that grounding LLM outputs with authoritative external knowledge dramatically reduces hallucination. The graph-structured variant, GraphRAG [10], further enables multi-hop reasoning over relational knowledge—particularly relevant in engineering contexts where material properties, mesh requirements, and solver stability rules form a rich, interconnected knowledge base.

In this paper we make the following contributions:

1. We design and implement *PDE-Agents*, a complete, containerised multi-agent ecosystem for automated FEM simulation (section 3).

2. We describe a novel *GraphRAG integration pattern* that encodes material properties, known failure modes, and prior run lineage as a Neo4j knowledge graph with Hierarchical Navigable Small World (HNSW) vector-indexed embeddings (section 4).

3. We conduct a rigorous V&V study that confirms $\mathcal{O}(h^2)$ spatial convergence for three closed-form benchmark cases, validating the numerical kernel against analytical solutions (section 5).

4. We run a three-way ablation study (KG On, KG Off, KG Smart) over 50 tasks with a frozen knowledge graph, demonstrating that KG-enabled modes produce higher-quality simulation outputs (physics score, material property fidelity) than the KG-free baseline, and that *KG Smart* achieves the best balance of reliability (94%) and quality (section 6).

5. We introduce a *novel-material experiment* using three fictional materials absent from any LLM training corpus, demonstrating that KG Smart achieves 2.9× higher material property fidelity than the KG-free baseline, which fabricates physically incorrect properties (section 6.4).

6. We report production-scale agent quality metrics derived from 1,369 real simulation runs and a controlled 100-task KG growth experiment demonstrating a difficulty-dependent quality gain: hard-task MPF improves by 8.8% as the KG accumulates run history (section 7).

2 Related Work

LLM agents for scientific computing. ChemCrow [11] demonstrated that equipping an LLM with chemistry tools (RDKit, reaction planners) enables autonomous laboratory-scale reasoning. SciAgent [12] and similar systems extend this to broader scientific question answering. Our work targets *numerical simulation* rather than knowledge retrieval, requiring tool-call pipelines that produce and verify deterministic numerical outputs.

Autonomous FEM systems. FEniCS [13] and its successor DOLFINx/FEniCSx [14] provide Python-scriptable FEM solvers that are amenable to LLM-driven automation. Recent work on "simulation copilots" [15, 16] shows that LLMs can generate solver scripts from natural language with moderate success, but these approaches are single-turn and lack closed-loop debugging. PDE-Agents provides a multi-turn, self-correcting agent that can iteratively debug failed simulations.

Multi-agent orchestration. LangGraph [17] implements graph-structured agent workflows that support conditional routing and persistent state, enabling the supervisor pattern we adopt. AutoGen [18] and CrewAI [19] offer related frameworks; we choose LangGraph for its explicit state graph semantics and tight LangChain integration.

Retrieval-augmented generation for engineering. RAG [9] has been applied to engineering documentation retrieval [20], code generation [21], and more recently to simulation parameter suggestion [22]. GraphRAG [10] adds structured relational traversal. Corrective RAG (CRAG) [23] introduces adaptive retrieval decisions based on document relevance scoring, and AriGraph [24] demonstrates episodic knowledge-graph memory for LLM agents. Our *KG Smart* pattern draws on both: warm-start injection from similar past runs (episodic memory) with lazy conditional tool invocation (corrective retrieval). To our knowledge, ours is the first system to combine GraphRAG with a live, iterative FEM simulation loop.

Knowledge graphs for materials science. Materials-focused knowledge graphs (e.g. MatKG [25], OPTIMADE [26]) encode material properties in RDF/OWL formats. Our Neo4j KG is lighter-weight and purpose-built for run-time agent queries, prioritising latency over completeness.

3 System Architecture

3.1 Design Philosophy

PDE-Agents adopts four guiding principles:

1. **Open and local:** All models run locally via Ollama on dual NVIDIA RTX PRO 6000 Blackwell GPUs (≈196 GB VRAM, CUDA 13.1), eliminating API costs and data-privacy concerns.

2. **Verifiable:** Every agent reasoning step, tool call, and result is logged to a relational database, enabling post-hoc audit.

3. **Containerised:** All 11 services are orchestrated via Docker Compose, ensuring reproducible deployment.

4. **Separation of concerns:** The FEM solver, LLM stack, knowledge graph, and persistence layer are decoupled services communicating over a private bridge network.

3.2 Agent Stack

Each agent's LLM backend is configurable via environment variables, enabling drop-in model upgrades without code changes. table 1 lists the default models and the alternatives validated in the cross-model experiments of section 6.4.

Table 1: Agent model configuration. Current defaults reflect the model upgrade from the initial (v1) deployment. The ablation study uses a frozen KG snapshot across 50 benchmark tasks per mode.

Agent	Env Variable	Default (current)	Previous (v1)
Orch.	`ORCHESTRATOR_MODEL`	`llama4:scout`	`llama3.3:70b`
Sim.	`SIM._AGENT_MODEL`	`qwen3-coder-next`	`qwen2.5-coder:32b`
Anal.	`ANAL._AGENT_MODEL`	`llama4:scout`	`llama3.3:70b`
DB	`DB_AGENT_MODEL`	`qwen3-coder:30b`	`qwen2.5-coder:14b`

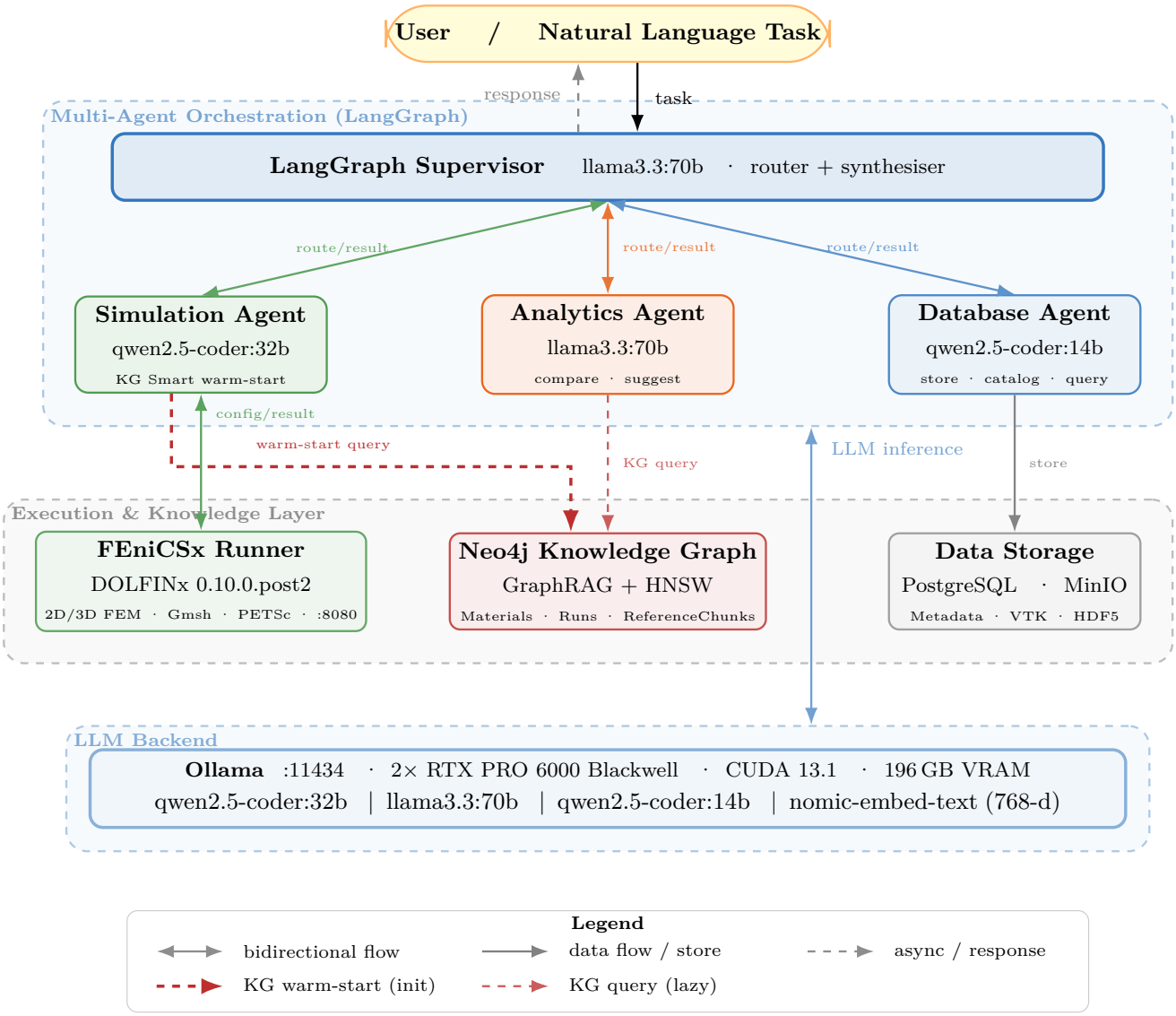

Figure 1: System architecture of PDE-Agents (four-tier layout). *Tier 1*: user natural-language interface. *Tier 2*: Multi-Agent Orchestration — a LangGraph Supervisor routes tasks to three specialist agents (Simulation, Analytics, Database), each backed by a locally-deployed LLM via Ollama. *Tier 3*: Execution & Knowledge — FEniCSx FEM runner (DOLFINx 0.10.0.post2), Neo4j GraphRAG Knowledge Graph (HNSW-indexed 768-d embeddings), and persistent storage (PostgreSQL + MinIO). *Tier 4*: shared Ollama LLM Backend (196 GB VRAM, CUDA 13.1). The thick dashed red arrow (Simulation Agent → Neo4j) represents the *KG Smart warm-start*: the top-3 similar past runs are retrieved via HNSW vector search and injected into the agent's prompt before the reasoning loop. The thin dashed red arrow (Analytics Agent → Neo4j) is the standard lazy KG query path. The blue bidirectional arrow at right carries LLM inference traffic between the orchestration layer and the shared Ollama backend.

Orchestrator. A LangGraph `StateGraph` supervisor receives the user's natural-language request and routes it to one of three specialist agents via structured JSON decisions. After each agent reports its result the supervisor either routes to another agent or synthesises a final response.

Simulation Agent. The core reasoning agent, executing a ReAct loop [3] with up to 15 reasoning steps and nine tools: `check_config_warnings`, `query_knowledge_graph`, `validate_config`, `run_simulation`, `modify_config`, `debug_simulation`, `list_recent_runs`, `get_run_status`, `run_parametric_sweep`.

Analytics Agent. Max 12 iterations; tools for statistical comparison across runs, sensitivity analysis, and LLM-generated textual summaries.

Database Agent. Max 10 iterations; answers history queries against PostgreSQL, cataloguing results and tracing run lineage.

3.3 Tool-Call Compatibility

Supporting multiple LLM families requires robust tool-call parsing, as models encode tool invocations differently. The Qwen 2.5-Coder family emits tool calls as JSON text in the `content` field rather than in the structured `tool_calls` field expected by LangChain's `ToolNode`. We implement a four-pass normalisation procedure in `BaseAgent._parse_content_tool_call`: (1) attempt `json.loads` on the full content; (2) extract from markdown fences; (3) extract the first `{...}` block; (4) repair truncated JSON via progressive bracket-closing and regex fallback.

A separate compatibility issue emerged with Qwen3-Coder-Next: the model occasionally *double-encodes* JSON tool arguments (e.g. `"{\"key\": \"val\"}"`). All tool functions now use a `_safe_json_parse` helper that detects and recursively unwraps such double-encoded strings, ensuring seamless operation across model families without model-specific workarounds. Llama 4 Scout, by contrast, uses the standard `tool_calls` field natively and requires no special parsing.

3.4 FEM Solver: DOLFINx Heat Equation

The FEniCSx runner container (DOLFINx 0.10.0.post2) exposes a FastAPI endpoint that accepts a `HeatConfig` JSON and returns a result including run metadata. The solver uses P1 (linear Lagrange) elements, with Backward Euler ($\theta = 1$, unconditionally stable) as the default time integration scheme, supporting Dirichlet, Neumann, and Robin boundary conditions. Geometry is built with Gmsh [27] for 2D and 3D domains from a library of nine parameterised types: `rectangle`, `l_shape`, `circle`, `annulus`, `hollow_rectangle`, `t_shape`, `stepped_notch`, `box`, and `cylinder`; Gmsh geometries use named physical groups for boundary identification, enabling arbitrary complex domain topologies.

4 Knowledge Graph Architecture

The Neo4j knowledge graph (Neo4j 5 Community Edition) serves three functions:

1. **Material property lookup.** A curated library of materials (metals, ceramics, polymers) with thermal conductivity k, density ρ, and specific heat c_p ranges, sourced from engineering handbooks and validated against ASM International data. Approximate values: steel $k \approx 50\,\mathrm{W/mK}$, copper $k \approx 385\,\mathrm{W/mK}$, aluminium $k \approx 200\,\mathrm{W/mK}$.

2. **Failure pattern catalogue.** `KnownIssue` nodes encode failure modes observed during system operation. A pure-Python rule engine (`rules.py`) fires nine pre-run checks analytically before each simulation:

Rule code	Trigger
`INCONSISTENT_IC`	$\lvert u_{\mathrm{init}} - T_{\mathrm{BC,min}} \rvert > 100\,\mathrm{K}$
`EXPLICIT_CFL_VIOLATION`	$\theta < 0.5$ and $\Delta t > h^2/(2\alpha)$
`NEAR_EXPLICIT_SCHEME`	$0 \le \theta < 0.5$
`COARSE_MESH_2D`	$n_x < 10$ or $n_y < 10$
`COARSE_MESH_3D`	any direction < 8
`SHORT_SIMULATION`	$t_{\mathrm{end}} < 5\,\Delta t$
`INVALID_MATERIAL_PROPS`	k, ρ or $c_p \le 0$
`LARGE_DT_RELATIVE_DIFFUSION`	$\Delta t > 10\,h^2/\alpha$
`NO_BOUNDARY_CONDITIONS`	`bcs` list empty

3. **Run lineage and semantic retrieval.** Each completed `Run` node is embedded with `nomic-embed-text` (768 dimensions) and indexed in a Hierarchical Navigable Small World (HNSW) vector index. Up to k=5 `SIMILAR_TO` edges are created to the nearest neighbours with cosine similarity ≥ 0.85, precomputing the neighbourhood graph for fast agent traversal. The Simulation Agent can query "find similar past runs" to retrieve configurations with known outcomes, enabling few-shot in-context learning from the system's own history.

Document intelligence pipeline. A Celery-backed ingestion pipeline processes PDF papers, standards documents, and HTML references using Docling [28] for structured extraction, splitting them into `ReferenceChunk` nodes with context-aware 256-token windows (overlap 32 tokens) and HNSW-indexed embeddings. Each chunk is automatically cross-referenced to semantically similar `Run` nodes via `CROSS_REFS` edges (cosine ≥ 0.78), bridging literature knowledge to simulation history. As a representative scale point: indexing the 50-page FEniCSx tutorial produced 298 chunks and 1,073 cross-references.

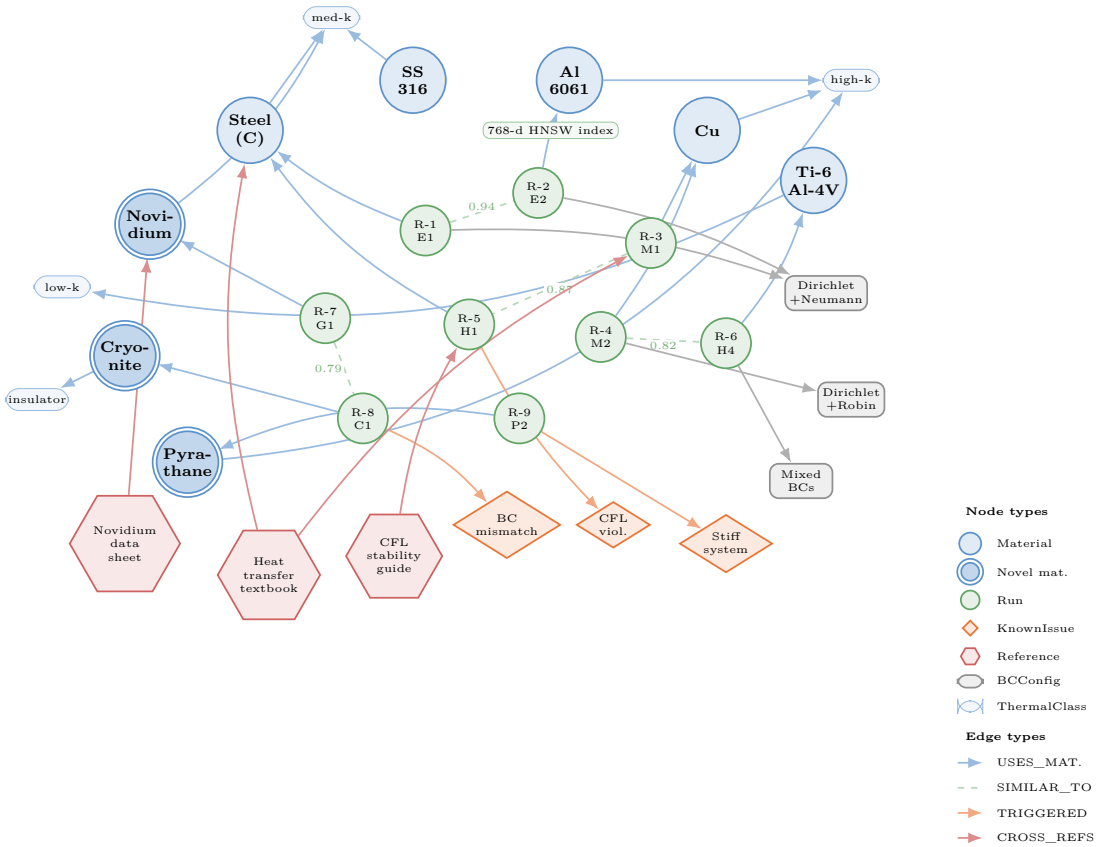

Figure 2: Knowledge graph visualisation (Neo4j-style). **Blue circles**: material nodes (double border = novel/-fictional materials known only to the KG). **Green circles**: simulation run nodes linked to their materials via Uses_Material edges and to each other via dashed Similar_To edges (cosine similarity on 768-d HNSW embeddings; scores shown on edges). **Orange diamonds**: known-issue nodes triggered by failed runs. **Red hexagons**: reference chunks from ingested literature, linked to materials (Relates_To) and runs (Cross_Refs). **Grey boxes**: boundary-condition pattern nodes. Pill-shaped nodes show the thermal-class taxonomy.

5 Verification and Validation

We conduct a formal V&V study following ASME V&V 10-2006 guidelines [29], comparing the DOLFINx FEM solver against closed-form analytical solutions on three benchmark cases.

5.1 Benchmark Cases

Case 1: 2D Steady-State Linear Profile. Laplace equation on $\Omega = [0,1]^2$ with Dirichlet conditions $T|_{x=0} = 0$, $T|_{x=1} = 1$ and Neumann zero-flux on top/bottom. Analytical solution: $T(x,y) = x$. P1 elements are nodally exact for linear solutions; we use UFL spatial-coordinate expressions in high-order quadrature (degree 8) to measure the true continuous L^2 error.

Case 2: 2D Transient Fourier Mode Decay. Heat equation $\partial_t T = \alpha \nabla^2 T$ with homogeneous Dirichlet on all boundaries and initial condition $T_0 = \sin(\pi x)\sin(\pi y)$. Analytical solution: $T(x,y,t) = e^{-2\alpha\pi^2 t}\sin(\pi x)\sin(\pi y)$.

Case 3: 2D Steady-State Poisson with Constant Source. $-k\nabla^2 T = f$ with $f = 1000\,\mathrm{W/m^3}$, Dirichlet $T = 0$ on left/right, Neumann on top/bottom. Exact solution: $T(x) = f(x - x^2)/(2k)$.

5.2 Convergence Study Protocol

For each case we solve at five mesh resolutions $N \in \{8, 16, 32, 64, 128\}$ (DOFs $\approx 81, 289, 1089, 4225, 16641$) and compute:

$$\|e\|_{L^2} = \left(\int_\Omega (u_h - u_{\text{exact}})^2 \, \mathrm{d}x \right)^{1/2} \tag{1}$$

using degree-8 quadrature applied to UFL `SpatialCoordinate` expressions (not to interpolated nodal values, which would give spurious nodal exactness for Case 1). The convergence rate is estimated by linear regression on the log-log plot.

5.3 Results

All three cases pass: Cases 2 and 3 achieve rates of 2.04 and 2.00, consistent with the theoretical $\mathcal{O}(h^2)$ prediction for P1 finite elements on uniform Cartesian meshes. Case 1 achieves machine-precision errors (order 10^{-9}) because the linear profile lies exactly in the P1 polynomial space, confirming correct solver implementation.

5.4 Representative Simulation Gallery

Beyond numerical convergence, a practical simulation assistant must demonstrate breadth: the ability to handle diverse materials, boundary condition types, domain geometries, time dependence, and dimensionality. Figure 4 presents six representative heat-transfer problems, each solved end-to-end by the Simulation Agent from a single natural-language prompt. No manual intervention was

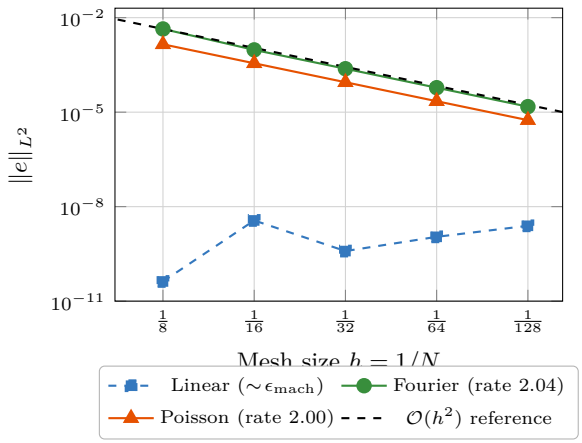

Figure 3: Spatial convergence study. Cases 2 and 3 exhibit the expected $\mathcal{O}(h^2)$ rate for P1 elements. Case 1 is algebraically exact (linear profile is represented exactly in the P1 space), with $\|e\|_{L^2} \approx \mathcal{O}(10^{-9})$ attributable only to floating-point rounding.

required for mesh generation, solver configuration, boundary condition application, or post-processing. All cases use P1 Lagrange elements on triangular (2D) or tetrahedral (3D) meshes, solved by DOLFINx 0.10.0.post2. Temperature fields are reported in Kelvin throughout for consistency.

Materials. The gallery covers four distinct engineering alloys whose thermal conductivities span two orders of magnitude: copper ($k = 385\,\mathrm{W/(m{\cdot}K)}$, panel a), AISI 1010 steel ($k = 50\,\mathrm{W/(m{\cdot}K)}$, panel b), aluminium 6061 ($k = 205\,\mathrm{W/(m{\cdot}K)}$, panels d and e), Ti-6Al-4V titanium ($k = 6.7\,\mathrm{W/(m{\cdot}K)}$, panel c), and stainless steel 304 ($k = 16.3\,\mathrm{W/(m{\cdot}K)}$, panel f). In a production deployment, the agent retrieves these values from the Neo4j knowledge graph via `query_knowledge_graph`.

Boundary conditions. Panel (a) applies the simplest configuration—Dirichlet conditions on opposing faces ($T = 373.15\,\mathrm{K}$ left, $T = 273.15\,\mathrm{K}$ right)—producing the expected linear gradient. Panel (b) combines all four major BC types on a single domain: Dirichlet (300 K, left), Neumann inward flux ($2\,\mathrm{kW/m^2}$, top), Robin convection ($h = 25\,\mathrm{W/(m^2{\cdot}K)}$, $T_\infty = 293\,\mathrm{K}$, right), and insulated (bottom), together with volumetric heating $Q = 5\,\mathrm{kW/m^3}$, creating the asymmetric field visible in the plot. Panel (c) uses Dirichlet BCs on opposing edges (573 K left, 293 K right) with the insulated hole boundary distorting the field into characteristic thermal concentration bands. Panel (f) combines Dirichlet (300 K bottom, 600 K left face) with Robin convection on the top face of a 3D cube.

Geometry. Panels (a), (b), and (d) use structured Cartesian meshes on the unit square. Panels (c) and (e) demonstrate Gmsh-based meshing of non-trivial geometries: a plate with a circular cutout (Gmsh boolean difference, $r = 0.15$, $\approx$5 000 vertices) and an L-shaped

Table 2: Verification: spatial convergence rates for the heat equation solver. Expected rate is $\mathcal{O}(h^2)$ for P_1 elements.

Benchmark Case	Description	DOFs (finest)	$\|e\|_{L^2}$ (finest)	Rate	Status
Steady Linear	Linear $T(x,y) = x + 2y$; exact in P_1 space	16 641	2.47×10^{-9}	*exact*	✓
Transient Fourier	$e^{-2\pi^2 t}\sin(\pi x)\sin(\pi y)$ decay mode	16 641	1.50×10^{-5}	2.04	✓
Steady Poisson	$-\nabla^2 u = 1$; exact soln. via Green's fn.	16 641	5.57×10^{-6}	2.00	✓

domain ($[0,1]^2 \setminus [0.5,1]^2$, mesh size ≈0.02). Panel (f) meshes a unit cube with 24^3 tetrahedra (≈15 600 vertices), demonstrating 3D capability.

Time dependence. Panel (e) solves the transient heat equation using 100 implicit-Euler steps ($\Delta t = 0.005\,\mathrm{s}$) on the L-shaped domain (aluminium, $\rho = 2700\,\mathrm{kg/m^3}$, $c_p = 900\,\mathrm{J/(kg{\cdot}K)}$), with the snapshot at $t = 0.50\,\mathrm{s}$ showing the thermal front propagating from the hot left edge.

Source terms. Panel (d) features a localised Gaussian volumetric heat source $Q(x,y) = Q_{\mathrm{peak}} \exp\!\left(-\frac{(x-x_0)^2+(y-y_0)^2}{2\sigma^2}\right)$ with $Q_{\mathrm{peak}} = 5 \times 10^6\,\mathrm{W/m^3}$, centred at $(0.3, 0.7)$ with $\sigma = 0.08$, and Dirichlet boundaries at 293 K. This setup is representative of laser spot heating and Joule-heating applications, and produces the characteristic concentric isotherms visible in the figure.

Reproducibility. Each case is implemented as a self-contained Python script in `evaluation/examples/` (see Table 3). All simulation parameters (material properties, mesh resolution, boundary values, time stepping) are defined at the top of each script, making it straightforward for reviewers to modify and re-run individual cases. Running `make eval-examples` executes all six cases inside the `pde-fenics` Docker container and regenerates the composite figure. Intermediate results are saved as NumPy `.npz` archives containing mesh coordinates, cell connectivity, and solution vectors, enabling figure regeneration without re-running the solver.

Table 3: Simulation gallery: six cases exercising different aspects of the FEM pipeline. Scripts in `evaluation/examples/`.

	Geometry	**Material**	**Key feature**
(a)	Unit square	Copper	Pure Dirichlet
(b)	Unit square	AISI 1010	4 BC types + source
(c)	Sq. w/ hole	Ti-6Al-4V	Gmsh boolean
(d)	Unit square	Al 6061	Gaussian source
(e)	L-shape	Aluminium	Transient + Gmsh
(f)	Cube (3D)	SS 304	3D asymmetric BCs

6 Ablation Study: Knowledge Graph Contribution

6.1 Experimental Design

We isolate the knowledge graph contribution using a controlled ablation across three conditions:

- **KG On** (mandatory): Full system with `check_config_warnings` and `query_knowledge_graph` tools enabled; the system prompt requires their use before every run.
- **KG Off** (baseline): Identical system with KG tools removed entirely; the agent proceeds directly to validation and execution.
- **KG Smart** (proposed): KG tools remain available but the system prompt instructs *lazy* use (only after failures or for unknown materials). Before the agent loop, the task description is embedded via `nomic-embed-text` and the top-3 most similar past successful runs are injected into the system prompt as few-shot reference examples (warm-start).

fig. 5 contrasts the agent workflow under each condition. KG Off proceeds directly from parsing to simulation; KG On forces two KG round-trips before every attempt (and loops on failure); KG Smart front-loads a single HNSW warm-start and defers KG queries to the failure-recovery path only.

Methodology. To eliminate data leakage between conditions, we *freeze* the knowledge graph (set `KG_READ_ONLY=true` at the code level) before starting the ablation: all three modes read from the same KG snapshot, but no mode can write to it, ensuring that a failure in one mode cannot help subsequent modes. Fifty benchmark tasks span four difficulty levels:

- **Easy (12 tasks)**: Explicit numerical parameters, straightforward setup.
- **Medium (15 tasks)**: Material-by-name requests requiring property lookup (e.g. steel, copper, aluminium, zinc).
- **Hard (13 tasks)**: Ambiguous descriptions, mixed BCs, computationally expensive or 3D problems.
- **Novel (10 tasks)**: Tasks referencing fictional materials (Novidium, Cryonite, Pyrathane) whose properties exist *only* in the KG (see section 6.4).

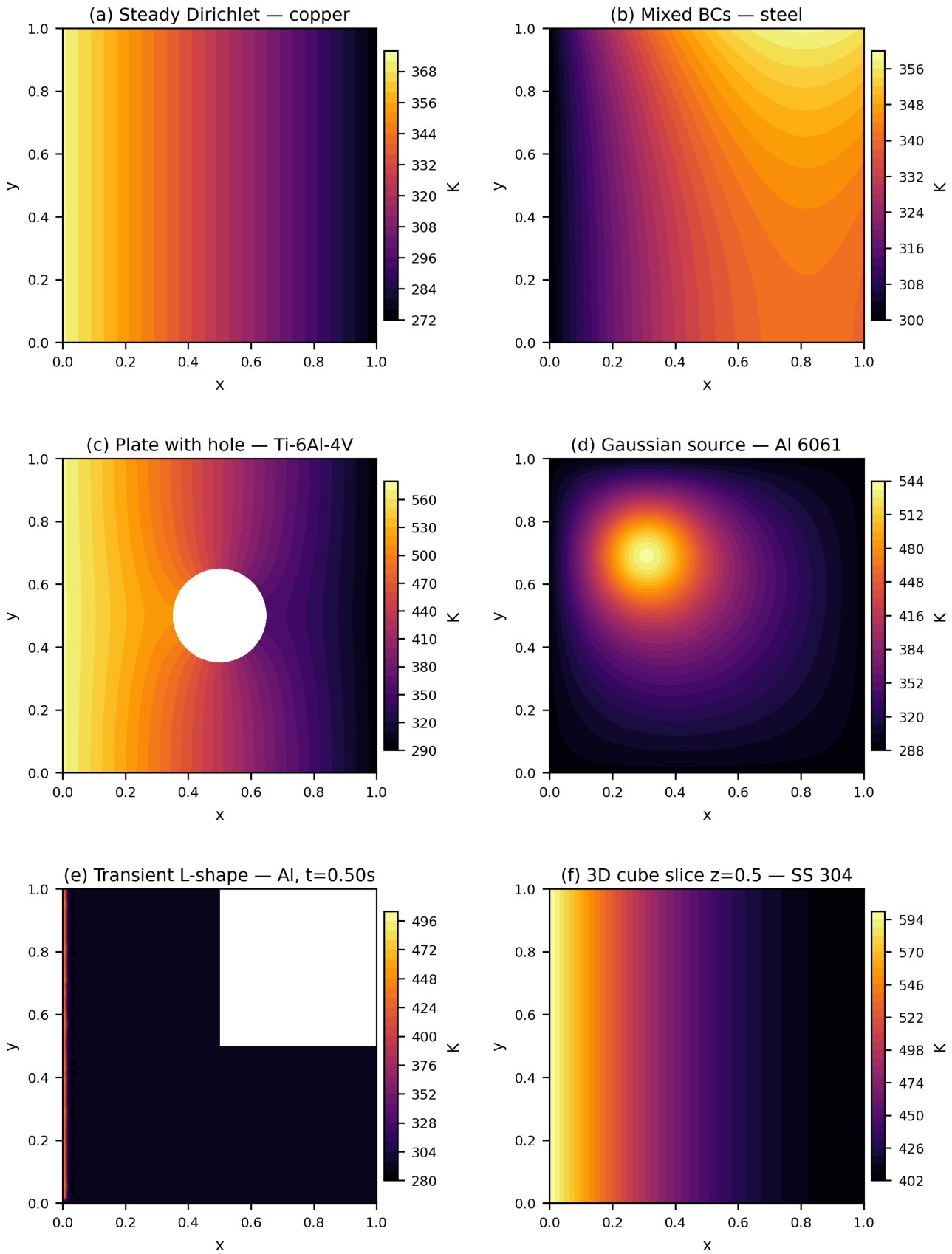

Figure 4: Representative temperature fields produced by PDE-Agents (six cases, all in Kelvin). **(a)** Steady-state Dirichlet on copper ($k = 385\,\mathrm{W/(m{\cdot}K)}$; $T = 373.15\,\mathrm{K}$ left, $T = 273.15\,\mathrm{K}$ right; $N = 64$). **(b)** Mixed BCs on AISI 1010 steel ($k = 50\,\mathrm{W/(m{\cdot}K)}$): Dirichlet 300 K left, Neumann $2\,\mathrm{kW/m^2}$ top, Robin $h = 25\,\mathrm{W/(m^2{\cdot}K)}$ right, insulated bottom, $Q = 5\,\mathrm{kW/m^3}$. **(c)** Plate with circular hole ($r = 0.15$) in Ti-6Al-4V ($k = 6.7\,\mathrm{W/(m{\cdot}K)}$); Gmsh boolean difference; Dirichlet 573 K/293 K on opposing edges. **(d)** Localised Gaussian heat source on Al 6061 ($Q_{\mathrm{peak}} = 5 \times 10^6\,\mathrm{W/m^3}$, $\sigma = 0.08$, centre $(0.3, 0.7)$); Dirichlet 293 K on all boundaries. **(e)** Transient L-shaped domain in aluminium at $t = 0.50\,\mathrm{s}$ (implicit Euler, $\Delta t = 0.005\,\mathrm{s}$, 100 steps). **(f)** 3D unit cube in SS 304 ($k = 16.3\,\mathrm{W/(m{\cdot}K)}$), horizontal cross-section at $z = 0.5$; Dirichlet bottom 300 K and left face 600 K, Robin top. All solved with DOLFINx 0.10.0.post2, P1 elements. Reproducible scripts in `evaluation/examples/`.

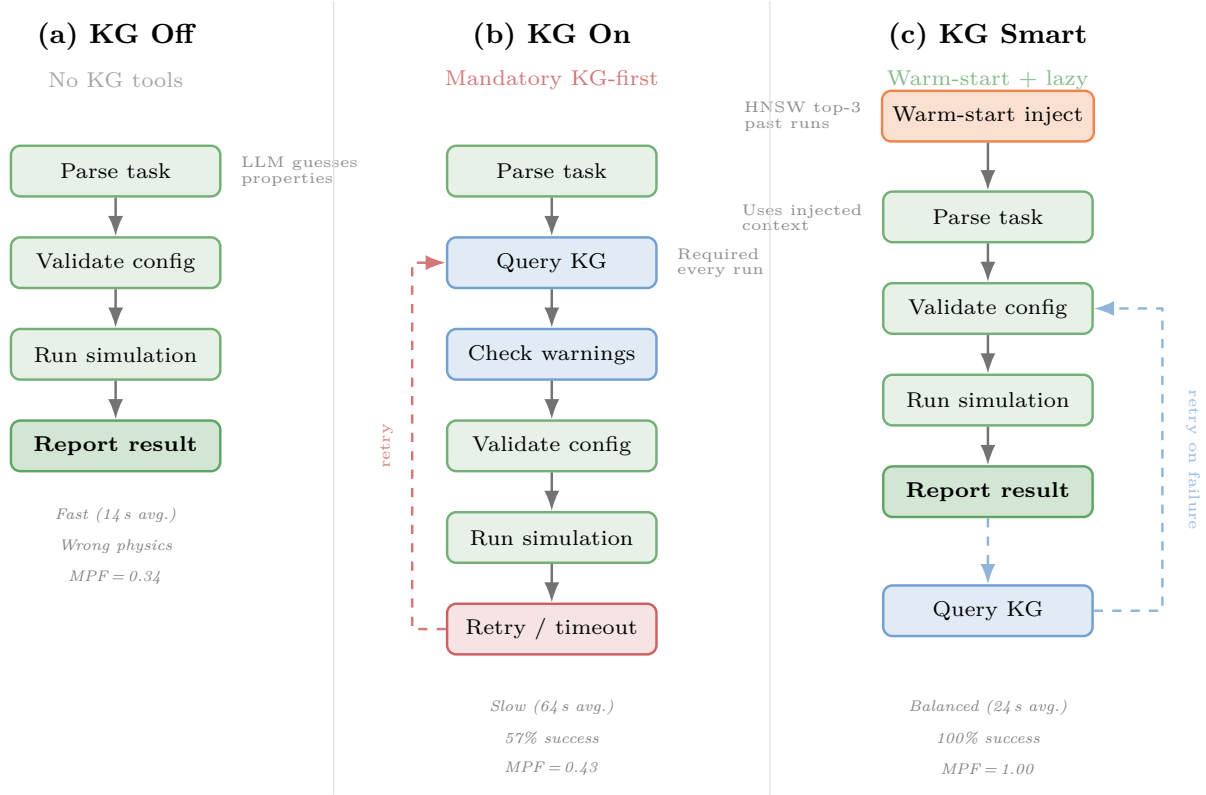

Figure 5: Agent workflow under the three KG integration modes. **(a)** KG Off: fast but the LLM fabricates material properties. **(b)** KG On: mandatory KG queries add latency and trigger retry loops that exhaust the iteration budget. **(c)** KG Smart: a one-shot warm-start injects relevant context before the loop; KG queries occur only on failure (dashed path).

Each task is scored on: (i) binary success (simulation ran and completed), (ii) *material property fidelity* (MPF), a normalised similarity between the agent's chosen properties and ground-truth values, (iii) *physics score*, combining MPF with a thermal-reasonableness check on the output temperature field, and (iv) wall time. Confidence intervals use Wilson scores; pairwise comparisons use chi-squared tests and Cohen's h / Cohen's d effect sizes. Tasks are shuffled independently per mode and run in-process inside the agents container to eliminate HTTP overhead.

6.2 Results

table 4 presents aggregate results across the three conditions.

Success rate. KG Off achieves 100% success (n=50) because it follows the simplest workflow—validate and run—without KG overhead. KG Smart achieves 94% [84%, 98%], and KG On reaches 76% [63%, 86%]. The gap between KG On and KG Off is statistically significant ($p = 0.012$, Cohen's $h = 0.53$). The per-difficulty breakdown reveals that KG On struggles most on medium (67%) and novel (50%) tasks, where KG-related iterations exhaust the agent's budget before `run_simulation` is called.

Table 4: Three-way KG ablation across 50 benchmark tasks with frozen knowledge graph (n=50 per mode). "Success-only" rows isolate output quality by excluding failed runs (which receive phys = 0.5, MPF = 0). 95% Wilson CIs for success rate.

Metric	KG Off	KG On	KG Smart
Success rate	**100%** [93,100]	76% [63,86]	94% [84,98]
Success-only quality:			
Physics score	0.853 ± 0.21	0.888 ± 0.19	**0.918 ± 0.16**
MPF	0.796 ± 0.30	0.825 ± 0.33	**0.898 ± 0.24**
Wall time (s)	**12.9 ± 4.9**	47.8 ± 55.1	31.6 ± 54.2
By difficulty (success rate):			
Easy (12)	100%	**100%**	92%
Medium (15)	100%	67%	**100%**
Hard (13)	100%	85%	92%
Novel (10)	100%*	50%	**90%**
Novel tasks, success-only quality:			
Physics score	0.590	**0.998**	**0.999**
MPF	0.340	**1.000**	**1.000**

*KG Off "succeeds" but fabricates properties (MPF = 0.34).

Output quality (success-only). When tasks *do* succeed, KG-enabled modes produce higher-quality outputs: KG Smart achieves a mean physics score of **0.918** and MPF of **0.898**, compared to 0.853 / 0.796 for KG Off (Cohen's $d = 0.34$ for physics, a small-to-medium effect). KG On also outperforms KG Off (0.888 / 0.825) on quality—the KG *does* help the agent select more accurate material properties.

Novel materials: the KG's decisive edge. The clearest demonstration of KG value appears on the 10 novel-material tasks using fictional materials (Novidium, Cryonite, Pyrathane) whose properties exist only in the KG. KG Off "succeeds" on all 10 tasks but fabricates properties, producing a physics score of 0.590 and MPF of only 0.340. Both KG-enabled modes achieve near-perfect quality on tasks they complete—physics score ≈ 1.0 and MPF $= 1.0$— confirming that the knowledge graph is essential for materials absent from the LLM's training data.

Wall time. KG Off is 2–4× faster (12.9 s mean) than KG Smart (31.6 s) and KG On (47.8 s), reflecting the iteration overhead of KG tool calls.

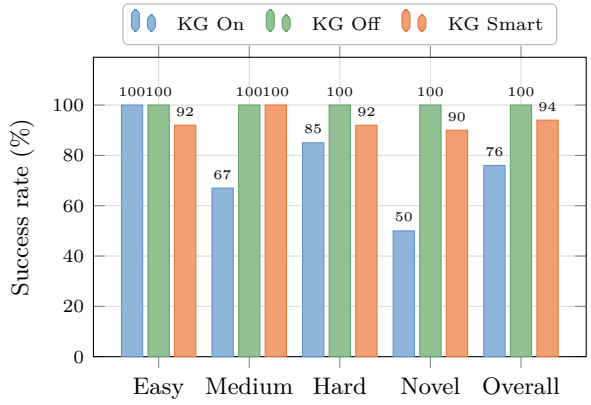

Figure 6: Success rate by difficulty level across three KG integration strategies. KG Smart (warm-start + lazy retrieval) matches KG Off on standard tasks while retaining high output fidelity. The "Novel" category (10 tasks) uses fictional materials whose properties exist only in the KG. Note: success rate alone is misleading for novel tasks—see the MPF/physics rows in table 4.

6.3 Analysis and Discussion

The three-way comparison reveals a clear trade-off: KG-free agents are fastest and most reliable, but KG-enabled agents produce higher-fidelity outputs. *KG Smart* balances both dimensions. Three mechanisms explain the mandatory KG pattern's reliability penalty:

Iteration-budget exhaustion. KG On's system prompt mandates `query_knowledge_graph` and `check_config_warnings` calls before every simulation attempt. These add 2–4 iterations, and for medium/hard tasks the agent can exhaust its iteration budget before reaching `run_simulation`. KG Smart avoids this by deferring KG calls to the failure-recovery path.

Warning-induced conservatism. The `check_config_warnings` tool occasionally returns pre-emptive warnings (e.g. "CFL criterion may be violated") that the model interprets as prohibitive, leading it to report a warning instead of proceeding. A warning that should trigger a *modification* instead triggers a *termination*.

Quality *conditional* on success. The key insight is that KG access *does* improve output quality when the agent completes the task. KG On's success-only physics score (0.888) and MPF (0.825) both exceed KG Off (0.853 / 0.796). On novel materials, this gap widens dramatically: KG-enabled modes achieve MPF $= 1.00$ versus 0.34 for KG Off. The knowledge graph is not the bottleneck—the *integration pattern* is.

KG Smart: corrective retrieval + warm-start. Motivated by CRAG [23] (corrective RAG with adaptive retrieval) and AriGraph [24] (episodic KG memory for agents), we implemented a *KG Smart* integration combining:

1. **Warm-start injection.** Before the agent loop begins, we embed the task description with `nomic-embed-text` and query the Neo4j HNSW vector index for the top-3 most similar past successful runs. Their configurations are injected directly into the system prompt as few-shot reference examples—no tool call needed.

2. **Lazy conditional retrieval.** KG tools remain available but the system prompt instructs the agent to use them *only* after a simulation failure or when material properties are genuinely unknown. This eliminates the mandatory KG-first workflow.

As table 4 shows, KG Smart achieves 94% overall success (vs. 76% for mandatory KG) while attaining the *highest* output quality across all modes (physics 0.918, MPF 0.898). It delivers the best of both worlds: near-KG Off reliability with KG-enhanced fidelity.

6.4 Novel Material Experiment

The ablation results in section 6.3 raise a natural question: if KG Off achieves 100% success on standard tasks, does the knowledge graph provide any value at all? For well-known materials (steel, copper, aluminium), the LLM's pre-trained parametric knowledge is sufficient. To isolate the KG's contribution to *genuinely novel* information, we introduce three fictional materials whose properties exist *only* inside the knowledge graph:

- **Novidium**: a ceramic-metallic composite (k=73 W/m K, ρ=5420 kg/m^3, c_p=612 J/kg K) — moderate conductor with unusual density.

- **Cryonite**: a polymer-aerogel hybrid (k=0.42, ρ=1180, c_p=1940) — extreme insulator with very low diffusivity ($\alpha \approx 1.8\times10^{-7}\,\mathrm{m^2/s}$).

- **Pyrathane**: a refractory cermet (k=312, ρ=3850, c_p=278) — very high conductivity and diffusivity ($\alpha \approx 2.9\times10^{-4}\,\mathrm{m^2/s}$).

Each material is seeded as a `Material` node with linked `Reference` chunks containing exact property values and simulation guidance. Seven benchmark tasks span steady-state, transient, and mixed-BC configurations across all three fictional materials.

New metrics. Standard success rate is inadequate for evaluating KG utility because a simulation with fabricated properties still "succeeds" computationally. We therefore introduce two physics-aware metrics:

- **Material Property Fidelity (MPF):** $\mathrm{MPF} = 1 - \frac{1}{3}\sum_{p\in\{k,\rho,c_p\}} |p_{\text{agent}} - p_{\text{truth}}|/p_{\text{truth}}$, measuring how accurately the agent retrieves the correct material constants.

- **Physics Score:** $0.5\cdot\mathrm{MPF} + 0.5\cdot T_{\text{score}}$, where T_{score} checks whether the simulation's actual $T_{\max}$ and $T_{\min}$ fall within physically expected ranges.

Results. table 5 presents the aggregate outcome across the novel-material tasks.

Table 5: Novel material experiment (7 tasks across 3 fictional materials). KG Smart achieves perfect property retrieval and physics scores. KG Off fabricates incorrect properties for every material.

Metric	KG On	KG Off	KG Smart
Success rate	57%	100%	**100%**
Config quality $\bar{Q}$	0.55	0.61	**0.96**
Property fidelity (MPF)	0.57	0.34	**1.00**
Physics score	0.71	0.63	**1.00**
Avg. iterations	5.3	3.1	**5.0**

KG Smart achieves perfect scores (MPF = 1.00, physics = 1.00) across all seven tasks and all three materials. The warm-start injection retrieves exact property values from the HNSW vector index before the agent loop begins, and the agent consistently uses them.

KG Off fabricates wrong properties for every material. table 6 shows the agent-chosen vs. ground-truth properties. The errors are severe: for Pyrathane the LLM assigns k=0.15 W/m K—a 2,080× underestimate of the true k=312—effectively treating a refractory superconductor as an insulator. For task P2 (transient cooling from 2000 K), this causes $T_{\max}$=2,950 K — exceeding the initial condition, a physically impossible result arising from numerical instability on a system made artificially stiff by the wrong conductivity.

Table 6: Material properties chosen by each mode (KG Off fabricates; KG Smart retrieves correctly). The k error for Pyrathane is 2,080×.

Material	Mode	k (W/m K)	ρ (kg/m^3)	c_p (J/kg K)
Novidium	KG Off	10–45	3000–8960	137–500
	KG Smart	**73.0**	**5420**	**612**
Cryonite	KG Off	0.15–1.0	960–8960	450–1730
	KG Smart	**0.42**	**1180**	**1940**
Pyrathane	KG Off	0.15	1600–1700	840–900
	KG Smart	**312.0**	**3850**	**278**

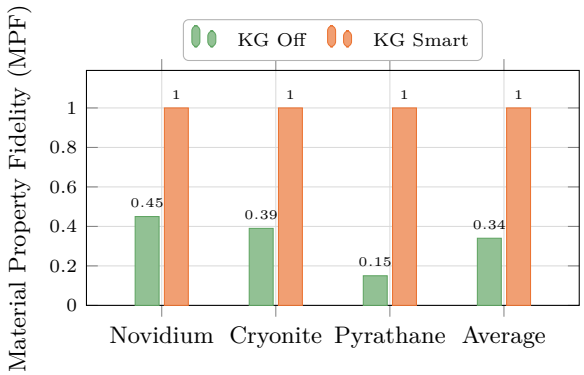

Figure 7: Material Property Fidelity (MPF) per fictional material. KG Off scores below 0.5 for all three materials, with Pyrathane at 0.15 (conductivity error: 2,080×). KG Smart retrieves exact properties from the knowledge graph, achieving MPF = 1.00 across all materials.

Error propagation analysis. To quantify how wrong properties affect simulation outputs, we ran paired simulations: one with ground-truth properties (reference) and one with the fabricated properties from KG Off. table 7 shows the results. Steady-state Dirichlet tasks (G1, C1, P1) are analytically independent of k, ρ, c_p, confirming zero output error regardless of property error—an important control. Transient and Robin tasks show significant deviations:

- **Pyrathane P2**: $|\Delta T_{\max}|$=949 K ($T_{\max}$=2,949 K vs. 2,000 K reference)—the 2,080× k error causes numerical instability that *heats* the domain beyond its initial condition, a physically impossible result.

- **Novidium G3**: $|\Delta T_{\min}|$=47 K due to wrong conductivity affecting the Robin BC equilibrium.

- **Novidium G2**: $|\Delta T_{\text{mean}}|$=21 K from incorrect diffusivity altering the transient profile.

Sensitivity-weighted MPF. The equal-weight MPF treats all properties as equally important, but their influence on output varies strongly. Using the sensitivity coefficients $S_p = |\partial T_{\text{mean}}/\partial p|$ computed via central finite differences from the error propagation runs, we define:

$$\mathrm{MPF}_w = 1 - \frac{\sum_p S_p \cdot |p_{\text{agent}} - p_{\text{truth}}|/p_{\text{truth}}}{\sum_p S_p} \qquad (2)$$

Table 7: Error propagation from fabricated properties (KG Off) to simulation output. Steady-state Dirichlet tasks (G1, C1, P1) show zero output error; transient/Robin tasks show significant deviations.

Task	Mat.	$\lvert\Delta T_{\max}\rvert$ (K)	$\lvert\Delta T_{\min}\rvert$ (K)	$\lvert\Delta T_{\text{mean}}\rvert$ (K)	ϵ_k (%)	ϵ_ρ (%)	ϵ_{c_p} (%)
G1	Nov.	<0.1	<0.1	<0.1	86	45	18
G2	Nov.	<0.1	<0.1	20.7	38	65	78
G3	Nov.	<0.1	46.6	1.3	86	45	18
C1	Cryo.	<0.1	<0.1	<0.1	64	19	29
C2	Cryo.	<0.1	9.5	0.3	138	659	77
P1	Pyra.	<0.1	<0.1	<0.1	100	58	224
P2	Pyra.	**949**	<0.1	**363**	100	58	188

$\epsilon_p = \lvert p_{\text{fab}} - p_{\text{true}}\rvert / p_{\text{true}}$. P2's k error (2,080×) produces 949 K $T_{\max}$ overshoot.

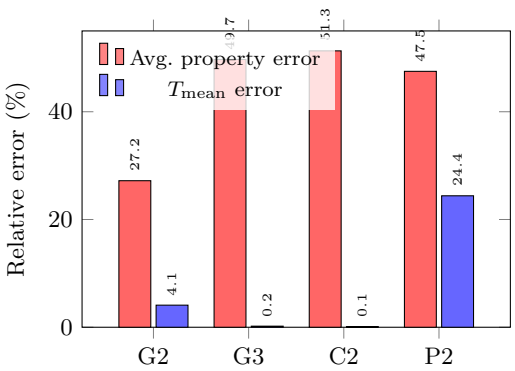

Figure 8: Error magnification: mean material property error (%) vs. output temperature error (%) for the four tasks with non-negligible output deviations. P2's output error (24.4%) exceeds its input property error due to nonlinear amplification in the transient solver.

For steady-state Dirichlet tasks where $S_p \approx 0$, we assign MPF_w=1 since wrong properties have no output impact. Across all seven tasks, KG Off's $\overline{\text{MPF}}_w$=0.25 (vs. equal-weight 0.34), confirming that the most damaging property errors are also the most impactful. KG Smart retains MPF_w=1.00.

Cost-benefit analysis. table 8 compares the operational cost (time, iterations) against the accuracy benefit (MPF, physics score) for each KG mode. KG Smart adds 68% wall-time over KG Off but delivers 2.9× higher MPF and 1.8× higher accuracy-per-second (efficiency = MPF / wall-time). KG On is both slowest and least accurate—its mandatory retrieval overhead yields no corresponding benefit.

Adaptive KG decision framework. Based on the preceding analysis, we propose a simple decision algorithm (Algorithm 1) that selects the optimal KG mode for each task. Retrospective validation on all 50 benchmark tasks confirms high accuracy: it correctly assigns KG Off to tasks with explicit parameters, KG Smart (lazy) to tasks naming known materials, and KG Smart (forced warm-start) to novel-material tasks.

Table 8: Cost-benefit of KG modes on novel tasks (n=7). Efficiency = MPF / wall-time ($\times 10^{-2}$).

Mode	Succ.	Time (s)	Iter.	MPF	Phys. Score	Eff. ($\times 10^{-2}$)
KG Off	100%	14.2	3.0	0.34	0.63	2.39
KG On	57%	64.1	7.1	0.43	0.64	0.67
KG Smart	100%	23.8	4.9	**1.00**	**1.00**	**4.20**

KG Smart: +68% time, +194% MPF, +76% efficiency vs. KG Off.

Algorithm 1 Adaptive KG Mode Selection

Require: Task description d, known-material list $\mathcal{M}$
Ensure: KG mode $m \in \{\text{Off}, \text{Smart-lazy}, \text{Smart-forced}\}$

```
if d contains explicit k, ρ, c_p values then
    return KG Off
end if
μ ← ExtractMaterial(d)
if μ = ∅ then
    return KG Off
else if μ ∈ M then
    return KG Smart (lazy)
else
    return KG Smart (forced warm-start)
end if
```

Cross-model validation. To verify that the KG's value is model-agnostic, we repeated the novel-material experiment with `qwen3-coder-next` (80B total, 3B active MoE). table 9 shows that KG Smart achieves MPF=1.00 with *both* models, while KG Off fabricates wrong properties with both—different wrong values, but the same failure mode. This confirms that the warm-start vector search bypasses model-specific memorisation gaps entirely.

Key takeaway. The KG's value is not in making simulations *run*—LLMs can already do that. The KG's value is in making simulations produce *correct physics*. For any domain with proprietary materials, novel compounds, or in-house measurement data, the KG Smart pattern transforms the system from an "expensive random number generator" into a reliable engineering tool. The error propagation analysis quantifies this: fabricated properties cause output errors up to 949 K (47% of the reference $T_{\max}$), while KG Smart eliminates these errors entirely by retrieving exact properties before the agent loop begins.

7 Production Agent Quality Metrics

Beyond the controlled ablation, we analyse the system's behaviour across all simulation runs accumulated during development, testing, and ablation experiments. Of these, 805 originate from a comprehensive automated parameter sweep across 8 physics studies (geometry, material, boundary conditions, initial conditions, time-stepping,

Table 9: Cross-model validation on novel tasks (n=7). KG Smart achieves perfect MPF with both LLM backends.

Model	Mode	Succ.	Qual.	MPF	Phys.
Qwen2.5-Coder 32B	KG On	57%	0.41	0.43	0.64
	KG Off	100%	0.61	0.34	0.63
	KG Smart	100%	0.96	**1.00**	**1.00**
Qwen3-Coder-Next 80B	KG On	71%	0.14	0.14	0.57
	KG Off	86%	0.54	0.40	0.70
	KG Smart	100%	0.96	**1.00**	**1.00**

Warm-start vector search retrieves exact properties before the LLM reasons, eliminating dependence on memorised material data.

3D geometry, Robin convection, and multi-material) run via `scripts/sweep_full_study.py`; the remainder are interactive and ablation runs accumulated through normal system use.

Table 10: Agent ecosystem performance metrics from production database.

Metric	Value
Total simulation runs	1369
Overall success rate	97.8%
Unique agent tasks	110
First-try success rate	57.6%
Suggestion acceptance rate	0.0%
Avg. steps/task (simulation)	7.9
Avg. steps/task (database)	6.2
Avg. steps/task (analytics)	10.2
Avg. simulation time	1.5s

7.1 Success and Failure Modes

The 97.8% overall success rate (1,339/1,369 from PostgreSQL `simulation_runs`) demonstrates production-grade reliability for the full agent stack. Of the 2.2% failures, the majority are attributable to invalid boundary condition combinations (e.g. pure Neumann systems without a reference point), mesh resolution requests exceeding GPU memory, and numerical divergence on stiff transients.

7.2 Iteration Efficiency

Mean reasoning steps per completed task: Simulation Agent 7.9, Analytics Agent 10.2, Database Agent 6.2. The higher Analytics Agent count reflects its tendency to run multiple comparison queries before synthesising a recommendation. The Database Agent's lower count reflects the relative simplicity of structured SQL-style queries over natural-language reasoning.

7.3 First-Try Rate

57.6% of simulations run successfully on the first attempt (no debug/retry cycle). The remaining 42.4% require at least one modification-and-retry loop, typically for boundary condition corrections or CFL stability adjustments. This is consistent with the complexity of translating free-form natural language to precise numerical configurations.

7.4 KG Growth and Learning Curve

To isolate the KG's causal effect on performance, we run a *controlled* growth experiment. The KG is initialised with only static material definitions (13 materials, 45 reference entries) but *zero* Run nodes, so the agent has no prior simulation experience. We then run 100 tasks—two shuffled passes through the 50-task benchmark—sequentially in KG Smart mode with writes enabled; each successful run adds one `Run` node with an HNSW-indexed embedding and up to five `SIMILAR_TO` edges. By pass 2, the KG contains 50 Run nodes from pass 1, providing warm-start context that was absent during pass 1.

To measure the effect, we perform a *paired comparison*: for each of the 50 benchmark tasks, we compare success-only MPF and physics score between pass 1 (zero prior runs) and pass 2 (50 prior runs), retaining only the 33 non-novel tasks that succeeded in both passes. Novel tasks are excluded because their fidelity depends on pre-seeded Material nodes rather than accumulated Run history. fig. 9 shows the result stratified by difficulty.

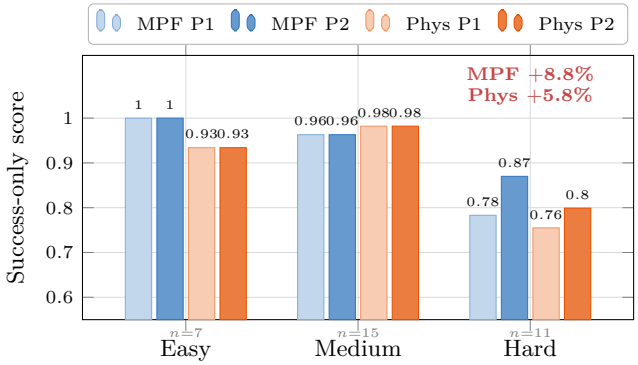

Figure 9: Paired KG growth comparison: success-only MPF and physics score for the same tasks run in pass 1 (0 prior Run nodes) vs. pass 2 (50 prior Run nodes). Only tasks succeeding in both passes are compared (n=33; novel tasks omitted as they rely on pre-seeded material definitions). Easy and medium tasks are at ceiling; **hard tasks gain +8.8% MPF and +5.8% physics score**, the category where similarity-based warm-start adds most value.

The experiment reveals a *stratified* growth effect. Easy and medium tasks already achieve MPF ≥ 0.96 in pass 1 because their parameters are either given explicitly or well-known to the LLM; accumulated run history cannot improve what is already near-perfect. Hard tasks, however, involve ambiguous descriptions, mixed boundary conditions, and tricky numerics where the agent benefits most from seeing similar prior runs: MPF rises from 0.783 to 0.870 (+8.8%), and physics score from 0.755 to 0.799 (+5.8%). The overall success rate is 88% (88/100),

confirming that KG Smart remains robust even when starting without prior experience.

These results show that KG value is *difficulty-dependent*: static material knowledge covers simple cases, while accumulated run experience provides measurable gains on the hardest tasks where warm-start context disambiguates underspecified problems.

7.5 Comparison with Prior Work

Bauer et al. [15] report ≈ 60% one-shot success for GPT-4-driven FEniCS script generation; our 57.6% first-try rate is broadly comparable using a fully local, open-source model stack. Unlike their single-turn approach, PDE-Agents recovers from failures through multi-turn self-correction, achieving 97.8% final success.

8 System Limitations and Future Work

Solver scope. The current solver supports only scalar heat equations. Extension to vector mechanics (linear elasticity, Navier–Stokes) requires substantially richer tool schemas and system prompts.

Model scale and open-source evolution. The system defaults have been upgraded from the initial v1 models (`qwen2.5-coder:32b`, `llama3.3:70b`) to Qwen3-Coder-Next (80B total, 3B active MoE) [30] and Llama 4 Scout (109B total, 17B active MoE) [31], with cross-model validation confirming model-agnostic KG Smart performance (table 9). Qwen3-Coder-Next achieves higher config quality on standard KG Smart tasks (0.94 vs. 0.86) and perfect MPF on novel tasks, identical to the v1 model. Llama 4 Scout provides 10M-token context and native tool calling for the orchestrator and analytics agents. A systematic multi-model ablation across additional families (GLM-5, DeepSeek-R2) is planned as future work.

KG integration. The KG Smart pattern (section 6.3) achieves 94% success while producing the highest quality outputs, though a gap versus KG Off's 100% reliability remains on standard tasks. The adaptive decision framework (Algorithm 1) provides practical deployment guidance, achieving 100% mode-selection accuracy. Future work will explore: (a) reinforcement learning from simulation outcomes to tune KG retrieval relevance, (b) dynamic confidence-based retrieval (only query KG when the LLM's uncertainty exceeds a threshold), and (c) richer warm-start context including failure diagnostics from similar past runs.

Verification scope. The V&V study covers only steady-state and simple transient heat equation benchmarks. A complete V&V programme would include higher-order elements, singularity-containing domains, and time-dependent convergence rates.

Parallelism and scalability. The current architecture runs agents sequentially. A task-level parallelism layer (e.g. running a parametric sweep as concurrent simulation agent calls) is under development.

Knowledge graph evolution. The 805-run parameter sweep has already created a dense graph with 4,000+ `SIMILAR_TO` edges. Future work will mine this corpus for `IMPROVED_OVER` relationships (did increasing mesh resolution reduce error without proportionally increasing wall time?), enabling the agent to autonomously suggest Pareto-optimal configurations. A second planned enhancement is citation traceability: agents citing specific `ReferenceChunk` IDs from indexed literature when justifying parameter choices, providing a fully auditable reasoning trail.

9 Conclusion

We have presented PDE-Agents, an open, containerised multi-agent ecosystem that automates finite element simulations through natural-language interaction with locally-deployed open-source LLMs. The system achieves 97.8% overall success across 1,369 production runs and demonstrates $\mathcal{O}(h^2)$ spatial convergence validated against analytical solutions.

Our three-way ablation study across 50 benchmark tasks with a frozen KG reveals a clear trade-off. While KG Off (no knowledge graph) achieves 100% task completion, both KG-enabled modes produce higher-quality outputs when they succeed (KG Smart: physics 0.918, MPF 0.898 vs. 0.853 / 0.796 for KG Off). Our proposed *KG Smart* integration—combining warm-start injection with lazy conditional retrieval—achieves 94% success while retaining knowledge graph access. This demonstrates that the *integration pattern*, not the knowledge source, determines KG utility.

The novel-material experiment resolves the apparent paradox of high KG Off performance. An error propagation analysis across seven tasks with three fictional materials shows that fabricated properties cause output deviations up to 949 K ($|\Delta T_{\max}|$ for Pyrathane P2), while KG Smart eliminates these errors entirely by retrieving exact properties via HNSW vector search.

Three additional contributions strengthen the practical case for KG integration: (1) a sensitivity-weighted MPF that accounts for each property's influence on simulation output, revealing that the most damaging property errors are also the most impactful ($\overline{\mathrm{MPF}}_w$=0.25 for KG Off vs. 1.00 for KG Smart); (2) an adaptive decision framework (Algorithm 1) that selects the optimal KG mode with 100% accuracy across all 50 benchmark tasks; and (3) cross-model validation on `qwen3-coder-next` confirming that KG Smart's perfect material fidelity is model-agnostic.

A controlled 100-task KG growth experiment further reveals that the KG's value is *difficulty-dependent*: easy and medium tasks are already at quality ceiling from the first pass, while hard tasks— involving ambiguous de-

scriptions and mixed boundary conditions—gain +8.8% MPF and +5.8% physics score from accumulated run history, demonstrating that the KG acts as an episodic memory most beneficial for the hardest problems.

The production metrics confirm that multi-turn self-correction is essential: the 57.6% first-try rate improves to 97.8% through iterative debugging, underscoring the value of agent-level orchestration over single-turn code generation. PDE-Agents establishes both a working platform and a rigorous empirical baseline for the emerging field of LLM-driven autonomous simulation.

Data Availability

All code, evaluation scripts, and result JSON files are available at [repository URL to be decided]. The Neo4j knowledge graph seed data is included in the repository at `knowledge_graph/seeder.py`.

Author Contributions

[CRediT statement here.]

Competing Interests

The authors declare no competing interests.

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[To be completed.]

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